

Reg. No															
---------	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

M.Tech DEGREE EXMINATION, JULY 2025

Second Semester

21CSC558J - DEEP LEARNING APPROACHES

(For the candidates admitted during the academic year 2024-2025)

Time: 3 Hours

Max. Marks: 100

(5 × 20 = 100 Marks)

Answer any 5 Questions

Marks BL CO

- | | | | |
|--|----|---|---|
| 1. Design a neural network in TensorFlow to perform multiclass classification of handwritten digits. Describe the network structure, activation functions, and how categorical cross-entropy is used. How does backpropagation update weights in this context? | 20 | 4 | 1 |
| 2. A team working on autonomous vehicles faces exploding gradient issues during model training. Explain how vanishing and exploding gradients occur and propose strategies using normalization (Batch/Group) and optimization (Adam, Adagrad). Include visual interpretation using loss curves. | 20 | 4 | 2 |
| 3. A company wants to develop a mobile app that classifies short video clips based on user activity (e.g., walking, running, jumping). Design a model using CNNs for spatial feature extraction and Bi-directional LSTM for sequence modeling. Explain how you would handle training difficulties with RNNs. | 20 | 4 | 3 |
| 4. How would you use transformers to perform sentiment classification on movie reviews? Outline the architecture using attention mechanisms and pretrained BERT embeddings. Discuss how fine-tuning improves performance over static word embeddings. | 20 | 4 | 4 |
| 5. You are developing a visual question answering (VQA) system for educational content. Describe how multi-modal data from images and natural language can be processed using CycleGAN and vision-language transformers. Illustrate the pipeline with intermediate output examples. | 20 | 4 | 5 |
| 6. A sentiment analysis task involves predicting mood trends over time based on diary entries. Combine TensorFlow models using embeddings, CNNs, and LSTM to build a hybrid text classification model. Justify the use of each component. | 20 | 4 | 1 |
| 7. A film production studio wants to generate cartoon-style animations from live-action footage. Explain how Pix2Pix or StyleGAN can be used for this task. Describe training the GAN, handling instability, and improving output realism using transformers or attention networks. | 20 | 4 | 3 |

* * * * *